

Yufeng (Alex) Xia

COMPUTER SCIENCE STUDENT · RESEARCH ASSISTANT

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Education

- Expected Grade: First class qualification. Expected Graduation: May 2025

Self Description

I am a self-motivated and detail-oriented researcher with a strong passion for exploring innovative solutions to complex problems in compiler technologies, decentralized learning, and embedded systems optimization. I enjoy conducting independent research, collaborating with multidisciplinary teams, and bridging theoretical knowledge with practical applications. My goal is to contribute to advancing technology through rigorous analysis, experimentation, and creative problem-solving.

Skills

Programming Languages

Advanced proficiency in C, c++, Python, Haskell, MIPS. Specialized in Java for network programming and troubleshooting.

Research Skills

Demonstrated ability to conduct independent and collaborative research, from problem identification to solution implementation. Experienced in designing experiments, analyzing results, and integrating theoretical knowledge with practical applications. Skilled in effectively communicating research findings through presentations and publications.

Development Tools

Proficient in using Git for version control, Maven for project management, and Docker for containerization, ensuring efficient development workflows.

Web Development

Proficient in building scalable and maintainable web applications using modern frameworks and technologies. Experienced in RESTful API design, microservices architecture, and containerized deployments with Docker and Kubernetes.

Cloud-Native Technologies

Hands on experience in cloud-native development, leveraging CI/CD pipelines, serverless computing, and infrastructure as code (IaC) for automated and efficient deployment of applications.

Compiler Technologies

Familiar with LLVM, GCC, LLVM Pass development, and compiler optimizations, including IR-level transformations and platform-specific tuning. Skilled in platform-specific tuning for embedded architectures. Experienced in debugging and profiling compiler-generated code to identify bottlenecks and improve runtime efficiency.

Machine Learning & AI

Experienced in decentralized and federated learning, with a focus on enhancing privacy and security in distributed systems. Skilled in using frameworks like PyTorch and TensorFlow for distributed model training and privacy-preserving applications. Actively engaged in virtual seminars and workshops led by leading academics

Linux

Experienced in the Linux environment with expertise in system administration, shell scripting, and environment configuration. Proficient in Linux kernel programming, including module development, debugging, and performance tuning for custom operating system functionality.

Experience

University of Edinburgh, Edinburgh

Edinburgh, UK

JUNIOR RESEARCH ASSISTANT

March 2024 - now

- Junior Research Assistant at Institute for Computing Systems Architecture (ICSA), under supervision of Dr. Antonio Barbalace.
- conducting research on compiler optimization techniques to improve performance and enable efficient code generation for embedded systems.
- Collaborating on cross-disciplinary projects involving compilers, and hardware-software co-design.
- Authoring technical reports, presenting research findings at internal meetings, and preparing manuscripts for academic publications in the domain of compilers and system architectures.

Senior Tech, Suzhou

Suzhou, China

BACK-END DEVELOPMENT INTERN AND SOFTWARE TESTER

June 2023 - September 2023

- Designed and implemented clean, maintainable, and scalable APIs for backend services, ensuring seamless data exchange and request handling.
- Participated in code reviews and debugging, working closely with developers to enhance system robustness and code quality.
- Modernized Uniqlo's delivery platform by transitioning from JFinal to Spring Boot, significantly improving system performance, scalability, and maintainability
- Contributed to the Uniqlo Android App testing, ensuring high usability.

Carnegie Mellon University - Decentralized Machine Learning Research Group

Remote

RESEARCH PARTICIPANT

Jan 2024 - May 2024

- Participated in cutting-edge research on decentralized machine learning, focusing on enhancing privacy and security in distributed systems.
- Participated in virtual seminars and workshops led by leading academics in the field, contributing to discussions on the latest developments and challenges in decentralized learning.
- Completed projects and case studies applying decentralized machine learning algorithms

Projects

Cosense Auto-Quantization based on sensor's specification

Edinburgh, UK

JUNIOR RESEARCH ASSISTANT

April 2024 - now

- CoSense is a compiler that uses sensor information from datasheet to help with optimization. It is developed based on LLVM.
- Currently developing and implementing an LLVM pass for the CoSense compiler, aiming to use sensor information from datasheets to optimize performance and power efficiency.
- Tech Stack: LLVM, Compiler, Compiler Optimization, Performance Benchmarking

Uniqlo Express Delivery Platform

Senior Tech, Suzhou

SOFTWARE ENGINEER INTERN

June 2023 - September 2023

- According to product requirements and development requirements, write easy maintenance and readable interface code for delivery platform system
- Work regularly with the development team on codereview to improve system code robustness
- Modernized and improved the efficiency of Uniqlo's express delivery platform by transitioning from JFinal to Spring Boot, enhancing performance and user experience.
- Refined business logic and introduced visual aids like process and Gantt charts for clear visualization of project progress.
- Tech Stack: Spring Boot, MySQL, Redis, Kafka, OpenResty, Documentation

Uniqlo Android App

Senior Tech, Suzhou

SOFTWARE ENGINEER INTERN

June 2023 - September 2023

- Contributed to enhancing the primary mobile touchpoint for Uniqlo's customers, focusing on rigorous testing for performance, user experience, and reliability.
- Proactively identified and resolved numerous bugs, significantly improving the app's usability and customer satisfaction.
- Skills: Documentation, Communication, Software Testing, Agile Development

PizzaDronz - Pizza Delivery Management Software

School of Informatics, University of
Edinburgh

PROJECT LEAD

September 2023 - December 2023

- Developed the system's order processing and validation logic, integrating complex algorithms to ensure accuracy and efficiency in order handling.
- Fine-tuned the A* pathfinding algorithm for drone navigation, optimizing delivery routes and avoiding no-fly zones.
- Designed and implemented a robust CI/CD pipeline using GitHub Actions for streamlined development and high code quality.
- Tech Stack: Java, Maven, RESTful services, Database, A* Pathfinding, CI/CD.

Hobbies

- **Linux Enthusiasm:** Deeply passionate about kernel development and a committed Linux user. I have a strong preference for Vim and enjoy using Arch and Fedora distributions. My hobbies include customizing development environments and delving into the complexities of Linux systems, contributing to open-source projects, and enhancing my understanding of kernel architecture.
- **Table Tennis:** An avid table tennis player, actively participating in school club activities and contributing to team dynamics and skill development.
- **Cycling and Fitness:** Enthusiastic about staying active and healthy through cycling and fitness routines, enjoying the challenge and refreshment it brings to my daily life.

Languages

Mandarin: Native Speaker | English: Fluent (IELTS 7.0 GRE 336)

Research and Publications

"Enhanced Pneumonia Detection in Chest X-Rays Based on Integrated Denoising Autoencoders and Convolutional Neural Networks" 2024

YUFENG XIA 2024

- Conduct research on integrating machine learning with medical diagnostics to improve cancer classification accuracy using denoising autoencoders.

"CoSense: Auto-Quantization Based on Sensor Specifications for Compiler Optimizations" Under Review

YUFENG XIA, PEI MU, ANTONIO BARBALACE 2025

- Developed a novel LLVM-based auto-quantization framework leveraging sensor specifications to enhance performance and reduce code size for embedded systems.
- Implemented quantization techniques tailored to sensor data, achieving significant improvements in execution speed and binary optimization.
- Evaluated CoSense on ARM Cortex-M0 and Cortex-A53 platforms, demonstrating measurable improvements in power, performance, and area (PPA).

Awards

2024 **Edinburgh Award (Work Experience)**, Recognition of my professional growth and the invaluable experiences gained as a Research Assistant University of Edinburgh

School Club Activities

2021-2024 **Table Tennis Enthusiast & Team Member**, Active participant in school club activities, contributing to team dynamics and skill development. Edinburgh

2021-2025 **BUCS Competitor**, Competed at a national level in the British Universities & Colleges Sport competitions. UK

2023 **Table Tennis Achievements**, Winner of Edinburgh Championship group stage, 3rd place in competition, MVP award, and "Most Improved Player in the league". Edinburgh

2024 **Table Tennis Achievements**, Secured 1st place in 2024 Edinburgh Championship Table Tennis Competition. Edinburgh